

Notes: Gabaix and Landier (QJE, 2008)

Why Has CEO Pay Increased So Much?

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1 Big picture

- **Question.** Can the dramatic rise in U.S. CEO compensation be explained by a **competitive market for managerial talent**, rather than mainly by rent extraction, weak governance, or the spread of incentive pay?
- **Setting.** The paper builds a **competitive assignment model** in which heterogeneous CEOs are matched to heterogeneous firms, and then confronts the model with **U.S. panel evidence**, **long-run U.S. time-series evidence**, and **cross-country evidence**.
- **Core challenge.** A convincing explanation has to account for three facts at once:
 1. the cross-section: larger firms pay more;
 2. the time series: CEO pay rose enormously after 1980;
 3. the international evidence: CEO pay differs across countries.

Existing theories often produce suggestive comparative statics, but not a **quantitative benchmark** that can explain all three dimensions together.

- **Main theoretical contribution.** The paper derives a closed-form **dual scaling equation** for CEO pay. In equilibrium, pay depends on:
 1. the CEO's **own firm size**, and
 2. the **aggregate size of large firms** in the economy.

The key technical step is to combine a Rosen–Terviö-style assignment model with **extreme value theory** for the spacing of top talents.

- **Main empirical contribution.** The paper shows that, for large U.S. firms, the model's core predictions fit the data surprisingly well:
 - **market capitalization** is the best proxy for firm size;
 - the cross-sectional elasticity of pay to own firm size is about **1/3**;
 - the time-series elasticity of pay to aggregate firm size is consistent with **constant returns to scale**;
 - a large fraction of cross-country compensation differences also lines up with firm size differences.
- **Why it matters.** The paper turns the debate over CEO pay into a quantitatively disciplined question. It gives a benchmark against which one can assess governance stories, skimming, incentive-based stories, and other explanations of top incomes.
- **Main empirical question.** Does the rise in firm size, especially the growth in the market value of large firms, quantitatively explain the rise in CEO pay?
- **Main results.**
 - The paper's central equilibrium equation predicts that CEO pay rises with both **own firm size** and the **size of the reference large firm**.
 - In U.S. panel regressions, the own-firm coefficient is around **0.3–0.4**, while the reference-firm coefficient is around **0.7**; together they imply $\gamma \approx 1$.

- Between 1980 and 2003, the average market value of the largest 500 U.S. firms rose by about a factor of **6**, and CEO pay also rose by about a factor of **6**.
 - In the calibration, the dispersion of top CEO talent is **very small**, but because talent is applied to huge firms, even tiny talent differences generate very large pay gaps.
 - Replacing CEO number 250 with CEO number 1 raises firm value by only about **0.016%**, yet CEO number 1 earns about **530%** more than CEO number 250.
 - Governance matters, but in the paper’s panel evidence the measured governance effect is **small** relative to the overall increase in pay.
 - The model also implies powerful **contagion effects**: overpayment at a small fraction of firms can raise compensation for all CEOs in general equilibrium.
- **Economic takeaway.** CEO pay can be understood as the equilibrium price of scarce managerial talent applied to very large corporate stakes. When the size of top firms increases, the market bids up the pay of top managers even if the distribution of talent changes little.

2 Data and measurement (key objects)

2.1 U.S. panel data on CEOs and firms

- **CEO pay data.** The main U.S. panel evidence comes from **ExecuComp** for **1992–2004**. The paper uses total compensation, TDC1, which includes salary, bonus, restricted stock granted, and the Black–Scholes value of stock options granted.
- **Firm universe.** Using **Compustat**, the paper selects each year the **top 500** or **top 1,000** firms by total market value.
- **Firm value measure.** Total firm value is measured as **equity market capitalization plus book value of debt**. The paper’s preferred proxy is therefore a broad market-value measure rather than accounting earnings alone.
- **Timing.** CEO compensation in year $t + 1$ is regressed on firm size measured at the end of fiscal year t . This avoids using a size measure observed after compensation is set.

2.2 Alternative firm-size proxies

- The paper compares three possible firm-size proxies:
 1. **total market value** (debt plus equity),
 2. **income** measured as EBIT,
 3. **sales**.
- The key conclusion is that **market capitalization** is the strongest predictor of pay and the only one that remains robustly positive when the different proxies compete in the same regression.
- **Interpretation.** This supports the paper’s preferred model in which CEO actions have a **persistent impact** on firm value, so that the relevant stake is the firm’s total market capitalization rather than one year’s earnings.

2.3 Long-run U.S. time-series objects

- The paper uses two compensation indices:
 1. the **Jensen–Murphy–Wruck** index, based on S&P 500 CEOs from 1970 onward;
 2. the **Frydman–Saks** index, which values compensation ex ante and provides earlier historical coverage.
- The paper compares these compensation series to the **mean market value of the top 500 firms** in Compustat, measured in real terms.
- These data are used to ask whether CEO pay and large-firm size move together over time.

2.4 Cross-country data

- International compensation data come from a **Towers Perrin** survey around 2000–2001.
- The paper measures country-specific reference firm size using **Compustat Global**, taking the **median net income of the top 50 firms** in each country.
- Because better international market-value data are not available in the same way, the cross-country exercise is more tentative than the U.S. exercise.
- The paper excludes **Belgium** in the main cross-country specification because it is viewed as too integrated with neighboring CEO labor markets.

2.5 Constructed theoretical objects

- **Firm rank n .** Lower n means a larger firm. $S(n)$ is firm size by rank.
- **Manager rank m .** Lower m means a more talented manager. $T(m)$ is managerial talent by rank.
- **Reference firm.** The benchmark rank is $n^* = 250$, the median firm in the universe of the top 500 firms.
- **Parameters.**
 - α : tail parameter of firm size distribution;
 - β : tail parameter of the upper talent spacing distribution;
 - γ : returns to scale of CEO impact with respect to firm size;
 - C : intensity with which talent affects firm value.

2.6 Unobservables and interpretation

- **Talent is unobserved.** The paper never directly observes CEO talent. It backs out the implications of talent differences from equilibrium wages and firm sizes.
- **Outside option.** The lowest-talent manager has reservation wage $w(N)$, but for top CEOs this outside option becomes negligible relative to pay.

- **Reference-size logic.** A CEO's pay depends not only on the size of his own firm but also on the overall demand for top talent, summarized by the size of the reference large firm.
- **Key measurement message.** Because the paper is about an *equilibrium market*, aggregate firm size matters even for explaining the pay of a single CEO.

3 Models and empirical specifications (all equations + interpretation)

3.1 A simple assignment framework

The paper starts from a firm with baseline earnings a_0 . CEO talent T raises earnings according to

$$a_1 = a_0(1 + CT).$$

The firm then chooses the CEO to maximize firm value net of compensation. In reduced form, the firm's problem can be written as

$$\max_T S + SCT - W(T),$$

where S is the relevant stake controlled by the CEO.

The paper then generalizes this to allow the CEO's impact to scale with firm size:

$$\max_T S + S^\gamma CT - W(T).$$

Interpretation.

- S is the size of the firm or the stake under managerial control.
- C measures how strongly talent affects firm performance.
- γ governs the returns to scale of CEO impact. If $\gamma = 1$, a 1% increase in firm size raises the value of CEO talent by 1%.

3.2 Competitive equilibrium and positive assortative matching

Let firm n have size $S(n)$ and manager m have talent $T(m)$. The firm's problem is

$$\max_m CS(n)^\gamma T(m) - w(m).$$

The first-order condition of the assignment problem implies

$$w'(n) = CS(n)^\gamma T'(n).$$

Interpretation.

- In equilibrium, better CEOs match with larger firms.
- The wage gradient for slightly better CEOs equals the marginal value of slightly higher talent at that firm's scale.
- This is the classic assignment logic: the market price of talent depends on where in the firm-size distribution that talent is used.

3.3 Firm size distribution and top-talent spacings

The paper assumes a Pareto firm-size distribution:

$$S(n) = An^{-\alpha}.$$

For the upper tail of talent, extreme value theory gives the spacing result

$$T'(x) = -Bx^{\beta-1},$$

or more generally $T'(x) = -Bx^{\beta-1}L(x)$ with $L(x)$ slowly varying.

Interpretation.

- α measures how fat the upper tail of firm size is. With $\alpha \approx 1$, firm size approximately follows **Zipf's law**.
- β measures how tightly packed top talents are near the very top.
- Extreme value theory is what lets the paper solve the model in closed form instead of leaving the talent distribution unspecified.

3.4 Equilibrium CEO pay: the dual scaling equation

Combining the assignment condition with the Pareto firm-size distribution and the upper-tail talent spacing gives

$$w(n) = \frac{A^\gamma BC}{\alpha\gamma - \beta} n^{-(\alpha\gamma - \beta)}.$$

The paper rewrites this in economically transparent form as the central equilibrium equation:

$$w(n) = D(n^*)S(n^*)^{\beta/\alpha}S(n)^{\gamma - \beta/\alpha},$$

where

$$D(n^*) = -\frac{Cn^*T'(n^*)}{\alpha\gamma - \beta}.$$

For the reference firm,

$$w(n^*) = D(n^*)S(n^*)^\gamma.$$

Interpretation.

- **Own-firm size effect.** Larger firms pay more because talent is more valuable when applied to larger stakes.
- **Aggregate-size effect.** When the size of top firms rises across the whole economy, every large firm must pay more to retain talent.
- **This is the market-equilibrium insight.** Pay is not determined only by a CEO's own firm, but by the *whole market demand* for top CEOs.

3.5 Cross-sectional, time-series, and cross-country implications

The dual scaling equation immediately implies three predictions:

1. **Cross-section:**

$$w \propto S^{\gamma-\beta/\alpha}.$$

This is the paper's explanation for **Roberts's law**, the empirical finding that CEO pay scales as a power of own firm size.

2. **Time series:** if all large firm sizes are multiplied by λ , then pay is multiplied by

$$\lambda^\gamma.$$

3. **Cross-country:** holding own firm size fixed, CEO pay across countries scales with the country-specific reference firm size as

$$S(n^*)^{\beta/\alpha},$$

provided national CEO labor markets are not fully integrated.

Interpretation.

- Cross-sectional elasticities are not the same as time-series elasticities.
- If $\gamma = 1$, then a sixfold rise in large-firm size implies a sixfold rise in CEO pay, even if the cross-sectional elasticity of pay to own firm size is only about 1/3.

3.6 U.S. panel regression

The paper takes the dual scaling equation to the data using the panel specification

$$\ln(w_{i,t+1}) = \ln D_i^* + \frac{\beta}{\alpha} \ln(S_{n^*,t}) + \left(\gamma - \frac{\beta}{\alpha}\right) \ln(S_{i,t}),$$

which is estimated as

$$\ln(w_{i,t+1}) = d + e \ln(S_{n^*,t}) + f \ln(S_{i,t}),$$

and, with industry effects,

$$\ln(w_{i,t+1}) = d_{\text{industry}} + e \ln(S_{n^*,t}) + f \ln(S_{i,t}).$$

Interpretation.

- e estimates β/α , the effect of **reference firm size**.
- f estimates $\gamma - \beta/\alpha$, the effect of **own firm size**.
- Therefore $e + f$ estimates γ , the time-series scaling parameter.

3.7 Time-series estimation of γ

For the long-run time series, the paper estimates

$$\hat{\gamma} = \frac{\ln(w_{2004}/w_{1970})}{\ln(S_{2003}/S_{1969})},$$

and also runs

$$\Delta_t \ln w_t = \hat{\gamma} \Delta_t \ln S_{t-1}.$$

Interpretation.

- The first formula asks: over the long run, how much did compensation rise relative to large-firm size?
- The second regression asks whether year-to-year changes in compensation line up with year-to-year changes in large-firm value.

3.8 Cross-country regression

For country i , the paper estimates

$$\ln w_i = c + \eta \ln S_{n^*,i}.$$

Interpretation.

- η should equal β/α if CEO labor markets are national and the model applies cleanly.
- This is a much noisier test than the U.S. panel and time-series tests, but it checks whether the same equilibrium logic has international bite.

3.9 Calibration of α , β , γ , and the magnitude of talent

The paper's benchmark calibration is:

$$\alpha \approx 1, \quad \gamma \approx 1, \quad \beta \approx \frac{2}{3}.$$

With $\beta > 0$, the upper tail of talent has an upper bound $T_{\max}$, and near the top,

$$\mathbb{P}(T > t) = B'(T_{\max} - t)^{1/\beta}.$$

To quantify how much top CEO talent matters, the paper uses

$$\left(\frac{\alpha\gamma}{\beta} - 1 \right) \left(1 - n_*^{-\beta} \right) \frac{w^*}{S(n^*)}$$

for the percentage increase in firm value from replacing the reference CEO with the top CEO.

Interpretation.

- The paper calibrates $n^* = 250$, $w^* = \$8.34$ million, and $S(n^*) = \$25$ billion.

- This yields the striking result that replacing CEO 250 with CEO 1 raises firm value by only about **0.016%**.
- Yet because even that tiny difference is applied to a huge firm, the top CEO can earn vastly more.

3.10 Extensions: heterogeneous firms, contagion, and top executives below the CEO

If firms differ in how sensitive they are to CEO talent, pay becomes

$$w = D(n^*) (\bar{C}^{1/\gamma} S(n^*))^{\beta/\alpha} (C^{1/\gamma} S)^{\gamma-\beta/\alpha},$$

with an average effective sensitivity term $\bar{C}$.

For contagion, if a fraction f of firms wants to pay λ times as much as similar firms, aggregate pay is multiplied by

$$\Lambda = \left[f \left(\frac{(1-f)\lambda}{1-\lambda f} \right)^{1/(\alpha\gamma-\beta)} + 1 - f \right]^{\alpha\gamma}.$$

For the top H executives within a firm,

$$w_{i,h} = D(n^*) \left(H^{-1} \sum_{k=1}^H C_k^{1/(\alpha\gamma)} \right)^{\beta} S(n^*)^{\beta/\alpha} S_i^{\gamma-\beta/\alpha} C_h^{1-\beta/(\alpha\gamma)}.$$

Interpretation.

- **Heterogeneous C .** Some firms are better able, or more eager, to convert talent into value.
- **Contagion.** A small set of firms overpaying can bid up compensation market-wide.
- **Top team.** The same equilibrium logic extends below the CEO to the top executives within the firm.

4 Identification and instruments (what makes the estimates credible)

4.1 This is not an IV paper

- The paper does **not** rely on a conventional causal design with an external instrument, natural experiment, or sharp policy discontinuity.
- Instead, it offers a **structural equilibrium benchmark**. The paper asks whether one coherent assignment model can simultaneously match:
 1. the cross-section of pay and firm size,
 2. the time series of pay and firm size,
 3. cross-country differences in pay.
- So the empirical strategy is best viewed as **disciplined model validation**, not a clean quasi-experimental estimate of a treatment effect.

4.2 What pins down the key parameters

- α is pinned down by the **upper-tail distribution of firm size**. The paper estimates it from the top 500 firms and finds approximate Zipf behavior.
- γ is pinned down by the **time-series response of CEO pay to aggregate firm size** and by the sum of coefficients in the panel regressions.
- β/α is pinned down by the **reference-firm-size coefficient** in the panel regressions and by the cross-sectional elasticity of pay to size.
- Once α and γ are near 1, the observed own-firm elasticity of about 1/3 implies $\beta \approx 2/3$.

4.3 Why market value is the crucial empirical object

- A central credibility check is the horse race among firm-size proxies. If the model is about the value of control over long-lived stakes, **market value** should outperform income and sales.
- Table I supports exactly that: market value is the most informative proxy for CEO pay.
- This matters because the main claim about the rise in pay from 1980 to 2003 is explicitly tied to the rise in **market capitalization of large firms**.

4.4 Why the paper's benchmark is persuasive

- The model gets the **cross-section** right: pay rises with own firm size.
- It also gets the **time series** right: pay rises roughly in line with aggregate large-firm value.
- It gets at least a noisy version of the **international pattern** right: countries with larger reference firms tend to pay CEOs more.
- Finally, the model's calibration leads to a coherent interpretation of superstar pay: **tiny talent differences** can generate **huge pay differences** when deployed at very large scale.

4.5 Why the paper is careful not to overclaim

- The paper explicitly says its preferred explanation is a **benchmark**, not the only plausible explanation.
- The benchmark does **not** fit the pre-1970 U.S. evidence well without amendment.
- Japan also looks difficult for the benchmark, suggesting that internal labor markets, social norms, or other institutions can weaken the simple market-size mechanism.
- Cross-country data are acknowledged to be **inferior** to U.S. data.
- Alternative explanations remain possible:
 - rising incentive pay,
 - weak governance,
 - changes in the CEO job,

- contagion from a small number of overpaying firms,
- expansion of outside sectors such as asset management,
- changes in the supply of talent.

4.6 Relation to the literature

- **Rosen (1981).** The paper is squarely in the economics-of-superstars tradition, but it makes the top-tail mathematics operational with extreme value theory.
- **Terviö (2003).** It is closest to Terviö’s competitive assignment view of the CEO market, but derives closed-form equilibrium pay rather than relying on a more implicit or semiparametric approach.
- **Governance and skimming literature.** The paper is deliberately a benchmark against rent-extraction stories, not a claim that governance never matters.
- **Top-income literature.** It links the growth of top incomes to the growth of the size of the stakes under the control of top talent.

5 Tables and figures

5.1 Table I: market capitalization is the best size proxy

- Table I compares **market capitalization**, **income**, and **sales** as predictors of CEO pay.
- When the proxies are entered together, market capitalization is the only one that remains robustly positive and significant.
- **How to read it.** This is not just a measurement choice. It tells the reader that the most relevant object for compensation is the **market value of the stake controlled by the CEO**, which is exactly what the theory emphasizes.
- **Punchline.** The paper’s preferred permanent-impact interpretation is empirically more compelling than the short-lived earnings-only view.

5.2 Table II: the dual scaling equation in the U.S. panel

- The table regresses log CEO pay on:
 1. log own firm market value,
 2. log market value of firm #250.
- In the baseline top-1000 specification, the coefficients are approximately:

$$\ln(\text{own market cap}) \approx 0.37, \quad \ln(\text{market cap of firm \#250}) \approx 0.72.$$

- **How to read it.**
 - The ≈ 0.37 coefficient is the cross-sectional own-firm elasticity.

TABLE I
CEO PAY AND DIFFERENT PROXIES FOR FIRM SIZE

	ln(Total compensation)			
	(1)	(2)	(3)	(4)
ln(Market cap)	0.34 (0.021) (0.021)	0.27 (0.008) (0.012)		
ln(Income)	0.006 (0.0138) (0.0149)		0.22 (0.008) (0.009)	
ln(Sales)	−0.08 (0.018) (0.020)			0.21 (0.008) (0.014)
Year fixed effects	Yes	Yes	Yes	Yes
Industry fixed effects	Yes	Yes	Yes	Yes
Observations	9,777	9,777	9,777	9,777
R ²	0.498	0.494	0.455	0.439

Explanation. We use ExecuComp data (1992–2004) and select for each year the 1,000 highest-paid CEOs, using the total compensation variable TDC1 at year t , which includes salary, bonus, restricted stock granted, and Black-Scholes value of stock-options granted. We regress the log of total compensation of the CEO in year t on the log of the firm's size proxies in year $t - 1$. All nominal quantities are converted to 2000 dollars using the GDP deflator of the Bureau of Economic Analysis. The industries are the Fama-French (1997) 48 sectors. To retrieve firm size information at year $t - 1$, we use Compustat Annual. The formula we use for total firm value (debt plus equity) is (data199*abs(data25)+data6-data60-data74). Income is measured as earnings before interest and taxes (EBIT), defined from Compustat as (data13-data14), and sales is measured as data12. We report standard errors clustered at the firm level (first line) and at the year level (second line).

TABLE II
PANEL EVIDENCE: CEO PAY, OWN FIRM SIZE, AND REFERENCE FIRM SIZE

	ln(Total compensation)							
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
	Top 1000				Top 500			
ln(Market cap)	0.37 (0.022) (0.016)	0.37 (0.020) (0.015)	0.37 (0.026) (0.015)	0.26 (0.056) (0.043)	0.38 (0.039) (0.020)	0.32 (0.039) (0.019)	0.33 (0.043) (0.026)	0.23 (0.074) (0.057)
ln(Market cap of firm #250)	0.72 (0.053) (0.066)	0.66 (0.054) (0.064)	0.68 (0.060) (0.061)	0.78 (0.052) (0.083)	0.73 (0.084) (0.089)	0.73 (0.085) (0.088)	0.74 (0.094) (0.081)	0.84 (0.080) (0.11)
GIM governance index			0.022 (0.010) (0.003)				0.023 (0.016) (0.007)	
Industry fixed effects	No	Yes	Yes	No	No	Yes	Yes	No
Firm fixed effects	No	No	No	Yes	No	No	No	Yes
Observations	7,936	7,936	6,393	7,936	4,156	4,156	3,474	4,156
R ²	0.23	0.29	0.32	0.60	0.20	0.29	0.32	0.63

Explanation. We use Compustat to retrieve firm size information at year $t - 1$. We select each year the top n ($n = 500, 1,000$) largest firms (in term of total market firm value, i.e., debt plus equity). The formula we use for total firm value is (data199*abs(data25)+data6-data60-data74). We then merge with ExecuComp data (1992–2004) and use the total compensation variable, TDC1 at year t , which includes salary, bonus, restricted stock granted and Black-Scholes value of stock options granted. All nominal quantities are converted into 2000 dollars using the GDP deflator of the Bureau of Economic Analysis. The industries are the Fama-French (1997) 48 sectors. The GIM governance index is the firm-level average of the Gomper-Ishi-Metrick (2003) measure of shareholder rights and takeover defenses over 1992–2004 at year $t - 1$. A high GIM means poor corporate governance. The standard deviation of the GIM index is 2.6 for the top 1000 firms. We regress the log of total compensation of the CEO in year t on the log of the market value (debt plus equity) in year $t - 1$, and the log of the 250th firm market value in year $t - 1$. We report standard errors clustered at the firm level (first line) and at the year level (second line).

- The ≈ 0.72 coefficient says aggregate firm size matters a lot for everyone's pay.
- The sum is close to 1, implying $\gamma \approx 1$.
- The governance index enters positively, but its magnitude is modest relative to the overall rise in CEO pay.
- **Punchline.** The paper's signature equilibrium prediction survives industry effects and even firm effects.

5.3 Figure I and Table III: the time series of CEO pay and firm size

TABLE I
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Explanation. We use ExecuComp data (1992–2004) and select for each year the 1,000 highest-paid CEOs, using the total compensation variable TDC1 at year t , which includes salary, bonus, restricted stock granted, and Black-Scholes value of stock-options granted. We regress the log of total compensation of the CEO in year t on the log of the firm's size proxies in year $t - 1$. All nominal quantities are converted to 2000 dollars using the GDP deflator of the Bureau of Economic Analysis. The industries are the Fama-French (1997) 48 sectors. To retrieve firm size information at year $t - 1$, we use Compustat Annual. The formula we use for total firm value (debt plus equity) is (data199*abs(data25)+data6-data60-data74). Income is measured as earnings before interest and taxes (EBIT), defined from Compustat as (data13-data14), and sales is measured as data12. We report standard errors clustered at the firm level (first line) and at the year level (second line).

TABLE III
CEO PAY AND THE SIZE OF LARGE FIRMS, 1970–2003

	$\Delta \ln(\text{Compensation})$	
	Jensen–Murphy–Wruck index	Frydman–Saks index
$\Delta \ln \text{Market}$	1.14 (0.28)	0.87 (0.30)
Constant	0.002 (0.032)	0.001 (0.033)
Observations	34	34
Adj. R ²	0.29	0.18

Explanation. We estimate for $t \geq 1971$

$$\Delta_t(\ln w_t) = \tilde{\gamma} \times \Delta_t \ln S_{t-1},$$

which gives a consistent estimate of γ . We show Newey–West standard errors in parentheses, allowing the error term to be autocorrelated for up to two lags. The Jensen, Murphy, and Wruck index is based on the data of Jensen, Murphy, and Wruck (2004). Their sample encompasses all CEOs included in the S&P 500, using data from Forbes and ExecuComp. CEO total pay includes cash pay, restricted stock, payouts from long-term pay programs, and the value of stock options granted, using after 1991 ExecuComp's modified Black-Scholes approach. Compensation prior to 1978 excludes option grants and is computed between 1978 and 1991 using the amounts realized from exercising stock options. The Frydman–Saks index is based on Frydman and Saks (2005). Total compensation is the sum of salaries, bonuses, long-term incentive payments, and the Black-Scholes value of options granted. The data are based on the three highest-paid officers in the largest 50 firms in 1940, 1960, and 1990. Size data for year t are based on the closing price of the previous fiscal year. The firm size variable is the mean of the biggest 500 firm asset market values in Compustat (the market value of equity plus the book value of debt). The formula we use is $\text{mktcap} = (\text{data199} \times \text{abs}(\text{data25}) + \text{data6} - \text{data60} - \text{data74})$. Quantities are deflated using the Bureau of Economic Analysis GDP deflator. Standard errors are in parentheses.

- Figure I plots the two compensation indices against the mean market value of the top 500 firms.

- The visual fact is the paper’s core macro result: **CEO pay and large-firm market value move together strongly after 1980.**
- Table III formalizes this by estimating the time-series scaling parameter γ , with values near **1**.
- The long-run ratio estimator gives:
 - $\hat{\gamma} = 1.17$ using Jensen–Murphy–Wruck,
 - $\hat{\gamma} = 0.85$ using Frydman–Saks,
 - $\hat{\gamma} = 1.01$ using the geometric-mean composite.
- **Punchline.** A sixfold increase in top-firm value maps naturally into a roughly sixfold increase in CEO pay.

5.4 Figure II and Table IV: cross-country evidence

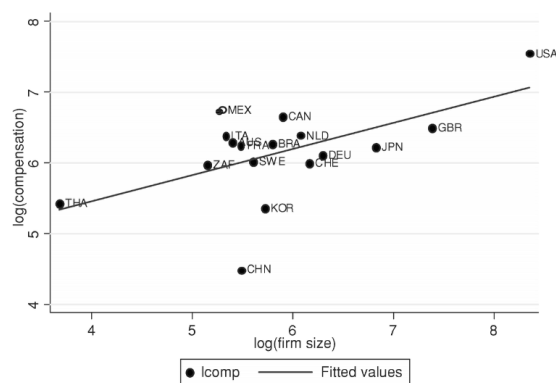


FIGURE II

CEO Compensation versus Firm Size across Countries

Notes. Compensation data are from Towers Perrin (2002). They represent the total dollar value of base salary, bonuses, and long-term compensation of the CEO of “a company incorporated in the indicated country with \$500 million in annual sales.” Firm size is the 2000 median net income of a country’s top 50 firms in Compustat Global.

TABLE IV
CEO PAY AND TYPICAL FIRM SIZE ACROSS COUNTRIES

	ln(Total compensation)			
	(1)	(2)	(3)	(4)
ln(median net income)	0.38 (0.10)	0.41 (0.098)	0.36 (0.096)	0.36 (0.12)
ln(pop)		−0.16 (0.092)		
ln(gdp/capita)			0.12 (0.067)	
“Social norm”				−0.018 (0.012)
Observations	17	17	17	17
R^2	0.48	0.57	0.58	0.52

Explanation. OLS estimates, standard errors in parentheses. Compensation information comes from Towers and Perrin data for 2000. We regress the log of CEO total compensation before tax in 1996 on the log of a country specific firm size measure. The firm size measure is based on 2001 Compustat Global data. We use the mean size of top 50 firms in each country, where size is proxied as net income (data32). The compensation variable is into U.S. dollars, and the size data are converted in U.S. dollars using the Compustat Global Currency data. The social norm variable is based on the World Value Survey’s E035 question in wave 2000, which gives the mean country sentiment toward the statement, “We need larger income differences as incentives for individual effort.” Its standard deviation is 10.4.

- Figure II plots CEO compensation against country-specific reference firm size.
- Table IV estimates the elasticity of CEO pay with respect to typical firm size across countries and finds a coefficient around **0.38**.
- **How to read it.** The cross-country result is weaker than the U.S. evidence, but it still points in the direction predicted by the theory: countries with larger firms tend to pay CEOs more.
- The paper is candid that the international data are noisy and incomplete, so this evidence is **suggestive**, not decisive.
- **Punchline.** The same basic assignment logic seems to travel internationally, but much less cleanly than in the U.S. evidence.

5.5 Figure III: Zipf’s law for top-firm market values

- Figure III shows the log–log plot of rank against firm market value for the top 500 firms in 2004.

- The fitted relationship is very close to linear, with an estimated slope implying $\alpha \approx 1$.
- **How to read it.** This figure is what makes the model analytically tractable and empirically grounded. The approximate Zipf law means the upper tail of firm size has a simple power-law form.
- **Punchline.** The empirical size distribution of large firms matches the paper’s theoretical assumption very closely.

TABLE III
CEO PAY AND THE SIZE OF LARGE FIRMS, 1970–2003

	$\Delta \ln(\text{Compensation})$	
	Jensen–Murphy–Wruck index	Frydman–Saks index
$\Delta \ln \text{Market}$	1.14 (0.28)	0.87 (0.30)
Constant	0.002 (0.032)	0.001 (0.033)
Observations	34	34
Adj. R^2	0.29	0.18

Explanation. We estimate for $t \geq 1971$

$$\Delta_t(\ln w_t) = \hat{\gamma} \times \Delta_t \ln S_{t-1},$$

which gives a consistent estimate of γ . We show Newey–West standard errors in parentheses, allowing the error term to be autocorrelated for up to two lags. The Jensen, Murphy, and Wruck index is based on the data of Jensen, Murphy, and Wruck (2004). Their sample encompasses all CEOs included in the S&P 500, using data from Forbes and ExecuComp. CEO total pay includes cash pay, restricted stock, payouts from long-term pay programs, and the value of stock options granted, using after 1991 ExecuComp’s modified Black–Scholes approach. Compensation prior to 1978 excludes option grants and is computed between 1978 and 1991 using the amounts realized from exercising stock options. The Frydman–Saks index is based on Frydman and Saks (2005). Total compensation is the sum of salaries, bonuses, long-term incentive payments, and the Black–Scholes value of options granted. The data are based on the three highest-paid officers in the largest 50 firms in 1940, 1960, and 1990. Size data for year t are based on the closing price of the previous fiscal year. The firm size variable is the mean of the biggest 500 firm asset market values in Compustat (the market value of equity plus the book value of debt). The formula we use is $\text{mkcap} = (\text{data199} \times \text{abs}(\text{data25}) + \text{data6} - \text{data60} - \text{data74})$. Quantities are deflated using the Bureau of Economic Analysis GDP deflator. Standard errors are in parentheses.

TABLE IV
CEO PAY AND TYPICAL FIRM SIZE ACROSS COUNTRIES

	$\ln(\text{Total compensation})$			
	(1)	(2)	(3)	(4)
$\ln(\text{median net income})$	0.38 (0.10)	0.41 (0.098)	0.36 (0.096)	0.36 (0.12)
$\ln(\text{pop})$		−0.16 (0.092)		
$\ln(\text{gdp/capita})$			0.12 (0.067)	
“Social norm”				−0.018 (0.012)
Observations	17	17	17	17
R^2	0.48	0.57	0.58	0.52

Explanation. OLS estimates, standard errors in parentheses. Compensation information comes from Towers and Perrin data for 2000. We regress the log of CEO total compensation before tax in 1996 on the log of a country specific firm size measure. The firm size measure is based on 2001 Compustat Global data. We use the mean size of top 50 firms in each country, where size is proxied as net income (data32). The compensation variable is into U.S. dollars, and the size data are converted in U.S. dollars using the Compustat Global Currency data. The social norm variable is based on the World Value Survey’s E035 question in wave 2000, which gives the mean country sentiment toward the statement, “We need larger income differences as incentives for individual effort.” Its standard deviation is 10.4.

5.6 Figure IV and the calibration of CEO talent

- Figure IV illustrates the implied shape of the upper tail of the talent distribution when $\beta \approx 2/3$: talent has an upper bound, and the density near that bound behaves like a square root.
- The paper then plugs in:
 - median top-500 CEO pay $w^* = \$8.34$ million,
 - reference firm value $S(n^*) = \$25$ billion,
 - $\alpha \approx 1$, $\beta \approx 2/3$, $\gamma \approx 1$.
- The implied value gain from replacing CEO 250 with CEO 1 is only about **0.016%**.
- Yet the implied pay gap is about **530%**.
- **Punchline.** This is the paper’s most memorable quantitative result: *small differences in talent can produce huge differences in pay when the stakes are huge.*

6 Short summary

- **Method.** Build a competitive assignment model of CEOs and firms, use extreme value theory to solve it in closed form, and then compare its predictions to cross-sectional, time-series, and cross-country data.

- **Identification logic.** No conventional instrument is used; instead, the paper disciplines the model with multiple empirical moments and key scaling relationships.
- **Calibration.** The benchmark values are $\alpha \approx 1$, $\gamma \approx 1$, and $\beta \approx 2/3$.
- **Main result.** The sixfold increase in the market value of large U.S. firms between 1980 and 2003 can explain the roughly sixfold increase in CEO pay in the same period.
- **Conceptual result.** CEO pay depends on both **own firm size** and **aggregate firm size**, because compensation is set in a market for top talent.
- **Quantitative result.** Differences in top CEO talent are inferred to be very small, but those small differences are magnified into very large compensation gaps by the scale of firms.

7 Conclusion

7.1 Core conclusion (what the paper establishes)

- The paper establishes a tractable competitive benchmark for CEO compensation.
- In this benchmark, the level of CEO pay is an equilibrium outcome of the assignment of slightly different talents to very different-sized firms.
- The model can explain a large part of the rise in U.S. CEO pay after 1980 through the growth in the market value of large firms, without requiring the entire increase to come from rent extraction or governance failure.

7.2 Economic intuition

- A slightly better CEO is not worth much in percentage terms.
- But when the firm is worth tens of billions of dollars, even a tiny percentage improvement is economically large.
- The largest firms compete most aggressively for the best CEOs, and that bidding pressure raises pay throughout the top of the market.
- So superstar compensation can arise from **tiny talent differences applied to huge stakes**.

7.3 Takeaway

This paper gives a clean way to think about CEO pay as an **equilibrium price in a market for talent**. Its main lesson is that the scale of top firms matters enormously. If the firms at the top become much bigger, the compensation of top managers can rise dramatically even without a dramatic change in governance or a dramatic increase in the dispersion of talent.